

TOPIC 26/60

What is Quantization?

Shrinking massive AI brains to fit in your pocket.

Ashish Jangra

linkedin.com/in/ashish-jangra



THE PRECISION PROBLEM

Most LLMs are trained in "High Definition" (FP16 or BF16). Each weight is stored as a 16-bit decimal number, which is very precise but takes up massive VRAM.

The Storage Cost:

A 70B parameter model in 16-bit precision needs ~140GB of VRAM. That's a professional \$10,000 GPU.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE QUANTIZATION SHIFT



Less Precision, Same Smarts

Quantization lowers the precision of those numbers. Instead of 16-bit, we map weights to 8-bit, 4-bit, or even 2-bit integers.

We don't need 16 decimal places of precision to know a weight is "important." A rounded estimate is often just as good.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE MATH

Rounding Down

Think of it like rounding π (3.14159...) to just 3.14. You lose a microscopic amount of accuracy, but you save massive storage space.

The Result:

We can compress a 140GB model down to 35GB (4-bit), allowing it to fit on a single consumer GPU.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE HARDWARE WIN

Local AI

Quantization turned AI from a "Cloud-only" luxury into something you can run on your MacBook or home PC.

Without quantization, your average consumer GPU would be too small to hold the model weights, making local LLM deployment impossible.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



KEY BENEFITS

Why Quantize?

It's the ultimate efficiency trade-off.

- **Memory:** Drastically reduced VRAM usage.
- **Speed:** Lower precision math (INT8) is often faster for hardware to compute.
- **Accessibility:** Enables "AI on the Edge" (phones, laptops).

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



COMMON FORMATS

GGUF

The Standard for CPUs

Highly popular for running models on Apple Silicon and generic RAM.

EXL2 / AWQ

The GPU Speedsters

Optimized for extreme inference speed on NVIDIA GPUs.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



THE TRADE-OFF

The Perplexity Hit

Quantization isn't free. If you squeeze too hard (e.g., to 2-bit), the model begins to "forget" knowledge and hallucinate.

The "Sweet Spot" is typically 4-bit (Q4_K_M). It's the point where you get a 4x reduction in size with almost zero noticeable impact on intelligence.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



SUMMARY OF PRECISION

FP16 (Original)

Perfect accuracy, massive hardware requirements.

Q4_K_M (Quantized)

The industry gold standard. 4x smaller, near-identical IQ.

Ashish Jangra

[linkedin.com/in/ashish-jangra](https://www.linkedin.com/in/ashish-jangra)



